Housing, borrowing, and family formation

A proposed repair of the illustrative model

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The most promising change is to distinguish *cash at purchase* from *earnings received during the period*. This permits a simple equilibrium in which young households rent small homes, save from their earnings, and buy larger homes when old. A planner choosing consumption and housing gives more housing to the young. With a further explicit restriction on child goods costs, the planner also chooses higher fertility.

This is a proposal for discussion. It preserves the two ages, logarithmic utility, child goods and space costs, the two upper housing limits, property taxes, and income–wealth heterogeneity. It changes the financial calendar, allows tenure to change with age, and omits ownership tastes in the first illustration. The main seminar deck has not been changed.

1. Environment. Households are young, then old. Young households choose completed fertility n. Their preferences remain

$$u^y = \log(c^y - \chi n) + \alpha \log(h^y - \kappa n) + \vartheta \log n, \qquad u^o = \log c^o + \gamma \log h^o + \omega_B \log e.$$

Lifetime utility is $u^y + \beta u^o$. The estate e includes the house's sale value at death. For the allocation comparison, set $\alpha = \gamma$. All preference and child-cost coefficients are positive.

Rental homes have size $h \leq H_R$ and owned homes have size $h \leq H_O$, where $H_R < H_O$. There is a common divisible stock $\bar{H}$. This retains the original common-stock interpretation: dwelling sizes and ownership can change, subject to these menus. It is not a fixed inventory of nonconvertible rental and owner units.

Cash and earnings. A young household has cash w_i at purchase, including every receipt already received. Its later working income is Y_i. These may differ across households. A mortgage finances at most the fraction ϕ of the purchase. It is repaid at the end of the young stage using working income and house-sale proceeds after current consumption and taxes are paid from liquid receipts; total wealth after repayment must be nonnegative. Retirement labor income is zero in the example below. Old households can sell their homes and choose either tenure.

Both current ages consume at the end of the same period. Old households leave their estates at that date. A young household's own old consumption comes one period later. Thus the dated planner compares goods and housing delivered on a common calendar.

Why this matters. A household can afford housing payments out of its earnings and still lack the deposit at purchase. Putting an entire long period's income in the bank before the purchase weakens that distinction. The proposal changes when income arrives; it does not exclude income that has already been received.

2. Household budgets. Let $q \in (0,1)$ be the price of a bond paying one unit at period end, P the stationary house price, and $\tau > 0$ the property-tax rate. Let $R = 1/q$ denote the gross bond return, and write $D = 1 - q + q\tau$. End-of-period rent per unit is DP/q. This is a small open economy with outside-sector bond and intermediary finance; warm-glow estates do not determine the next entrant's cash. If each age has mass N, the common upfront tax rebate is

$$T = \frac{q\tau P\bar{H}}{2N}.$$

The government finances this rebate against the period-end property-tax receipts. It introduces no additional goods.

A young owner chooses consumption, housing, and next-period net financial position a', measured immediately before the boundary sale of the house. The choice is subject to

$$\boxed{\begin{aligned} &qc^y + qa' + (1+q\tau)Ph^y = w_i + T + qY_i, \\ &\qquad (1-\phi)Ph^y \le w_i + T, \qquad a' + Ph^y \ge 0, \\ &\qquad qa' + \phi Ph^y \ge 0. \end{aligned}}$$

The deposit requirement says cash covers 20% when $\phi = .8$. The condition $qa' + \phi Ph^y \ge 0$ permits current consumption and property tax before sale; $a' + Ph^y \ge 0$ requires nonnegative wealth after repayment. A young renter instead has

$$qc^y + qa' + DPh^y = w_i + T + qY_i, \qquad a' \ge 0, \quad h^y \le H_R.$$

This timing requires the home to be chosen before the later earnings arrive. A purchase after that receipt would be a subsequent housing decision. At either age, current consumption and property tax are paid from liquid receipts before the house is liquidated. For a young household, house-sale proceeds may then repay the mortgage, and the next old-entry rebate is credited after settlement. For an old household, sale proceeds form the estate after current consumption and taxes have been paid. An old household may sell and repurchase under the unchanged old no-borrowing problem.

An old household's initial resources include its financial wealth, the house carried from youth, and the rebate:

$$Z = a' + Ph^y \mathbf{1}\{\text{previously owner}\} + T.$$

An old owner has budget and estate

$$qc^o + qe + DPh^o = Z, \qquad e = a^e + Ph^o, \qquad a^e \ge 0.$$

Here a^e is the financial estate delivered at death. An old renter has the same consolidated budget, with $e = a^e$ and $h^o \le H_R$. The owner's financial lower bound implies $e \ge Ph^o$. Old households need not leave liquid assets in the equilibrium used below.

The deposit can exclude large homes without an owner minimum. If

$$\frac{w_i + T}{(1-\phi)P} < H_R,$$

every home the young household can buy is smaller than the largest rental. A renter can replicate such an owner's consumption and future total wealth by increasing financial saving by the house's sale value. With mobile future tenure and no ownership taste, buying that smaller home has no advantage. If rental demand reaches H_R, the young household rents there and would like more space. This is the joint role of the deposit and rental-size constraints.

3. An equilibrium with the required allocation. First hold fertility fixed, with $0 < \kappa n_i < H_R$ and $\nu \overline{n} = 1$. Normalize each age mass to one and write $r = H_R$, $H = \bar{H}/N$, and $m = H - r$. The proposed equilibrium has all young households renting r and mean old housing $m > r$.

Define the old utility weight and housing coefficient by

$$K = 1 + \gamma + \omega_B, \qquad a = \min\left\{\frac{\gamma}{D}, \frac{\gamma + \omega_B}{1 + q\tau}\right\}.$$

The two terms cover positive financial estates and a binding estate floor. In either case, an uncapped old owner has

$$c_i^o = \frac{Z_i}{qK}, \qquad h_i^o = \frac{aZ_i}{KP}.$$

For young net resources $b_i = w_i/q + Y_i - \chi n_i$, set $G = 1 + \beta K$. The equilibrium price and rebate are explicit:

$$\boxed{\begin{aligned} J &= Gm + a\beta\left[\frac{D}{q}r - \frac{(1+q)\tau H}{2}\right], \\ P &= \frac{a\beta\overline{b}}{J}, \qquad T = \frac{q\tau PH}{2}. \end{aligned}}$$

The individual allocation is then

$$\boxed{\begin{aligned} x_i \equiv c_i^y - \chi n_i &= \frac{b_i + (1+q)T/q - (DP/q)r}{G}, \\ a_i' = \beta K x_i - T, &\qquad c_i^o = \frac{\beta}{q}x_i, \qquad h_i^o = \frac{a\beta x_i}{P}. \end{aligned}}$$

The next page states every inequality needed to verify this equilibrium. They are functions of primitives because P and T are already given by the displayed formulas.

A finite region where all the inequalities hold. One analytical family has

$$\begin{gathered} q = .5, \quad \phi = .8, \quad r = 1, \quad H = 2.5, \quad H_O = 2, \\ \alpha = \gamma = 1, \quad \omega_B = .25, \quad .3 \le \beta \le 1, \quad .01 \le \tau \le .02, \\ \left|b_i/\overline{b} - 1\right| \le .05, \qquad \overline{b} \ge 45 \max_i w_i. \end{gathered}$$

Every young household rents one unit and saves; each old household owns between 1.405 and 1.595 units. The old financial estate is zero. These conclusions hold throughout the stated region, by analytical inequalities. The numbers do not stand in for a numerical equilibrium proof.

The substantive restriction is low cash relative to later stage income. This is a sufficient example, with limited resource dispersion and no tenure tastes, not an empirical assessment of the parameter region.

4. Complete verification of the equilibrium statement. Let $b_e = \max\{\omega_B, qa\}$ and $G_R = 1 + \beta(1+\omega_B)$. For $t > 1$, define the utility advantage of the owner continuation by

$$\Delta(t) = \beta\gamma \log t + \beta\omega_B \log(b_e/\omega_B) - G_R \log\left(\frac{G - a\beta D/t}{G_R}\right).$$

This compares the optimized lifetime plan ending as an owner with the optimized plan renting in both ages; it is not a marginal-utility assumption.

Proposition. Suppose $J > 0$, $\bar{b} > 0$, and, at the explicit P, T, x_i, h_i^o above, every type satisfies

$\dfrac{w_i + T}{(1-\phi)P} < r,$	purchase cash limits the young;
$x_i > 0, \quad \beta K x_i > T,$	consumption and saving are positive;
$\alpha x_i > \dfrac{DP}{q}(r - \kappa n_i),$	the young rental limit binds;
$r < h_i^o < H_O,$	the old owner choice is feasible;
$\Delta(h_i^o/r) > 0,$	owning when old is preferred.

Then the displayed prices and allocations form a stationary competitive equilibrium. All young rent at their size limit, save, and buy larger homes when old. Prices are unique within this regime. Other equilibrium regimes have not been excluded.

Proof. With a future owner, the old value is $K \log Z$ plus a constant. The young problem is strictly concave; its saving condition gives $Z_i = \beta K x_i$. The stated cap and feasibility inequalities verify its optimum. The old estate floor is included in that value, even when it binds.

The alternative lifetime renter problem is jointly concave. Both rental limits bind under the displayed restrictions, and its adult consumption is

$$x_i^{RR} = \frac{Gx_i - DPr}{G_R}.$$

Subtracting its maximized value gives $\Delta(h_i^o/r)$. The deposit replication on page 2 rules out a better plan starting as an owner. Finally, the price formula gives $\overline{h}^o = m$, and $2T = q\tau PH$ balances the fiscal account. Rental intermediaries earn the bond return and all titles are accounted for.

Chosen fertility. At the rental limit, a private choice of fertility solves

$$\frac{\vartheta}{n_i} = \frac{\chi}{x_i} + \frac{\alpha\kappa}{r - \kappa n_i}.$$

Substitution for x_i gives a quadratic. The supporting calculation extends the displayed family by imposing small child goods costs and verifying tenure preference over *all* feasible fertility choices. A one-dimensional monotone equation then imposes replacement fertility. Heterogeneous fertility need not be replaced by a representative household.

5. The full dated planner. The planner reallocates consumption and housing among the households alive at the start of a period. First hold their fertility fixed. It chooses current tenure as well as home size, respects the housing stock and the size menus, and can relax private financial constraints. Individual young continuation wealth and old estates stay fixed. Future prices and opportunity sets are held fixed in this dated comparison. Thus their future utility terms are constant in the planner's problem.

Let $\mathcal{C}$ be total goods delivered to the two current ages, per unit of young-cohort mass. With $\alpha = \gamma$, its problem is

$$\max_{c^y, c^o, h^y, h^o, \text{tenure}} \int_y [\log(c_i^y - \chi n_i) + \alpha \log(h_i^y - \kappa n_i)] + \int_o \left[\log c_j^o + \alpha \log h_j^o\right]$$
$$\text{subject to} \quad \int_y c_i^y + \int_o c_j^o = \mathcal{C}, \qquad \int_y h_i^y + \int_o h_j^o = H,$$

with positive adult consumption and the stated size menus. Since ownership has no direct taste cost, the planner can assign any household a feasible owned home up to H_O. This permission to change tenure is essential: a young household whose rental tenure were fixed could not exceed H_R.

The solution equalizes adult consumption. With slack planner housing caps,

$$X = \frac{\overline{x}^y + \overline{c}^o}{2}, \qquad s = \frac{H - \kappa \overline{n}}{2}, \qquad h_i^{y,F} = s + \kappa n_i, \quad h_j^{o,F} = s.$$

With binding caps, replace these housing choices by $\min\{H_O, s + \kappa n_i\}$ and $\min\{H_O, s\}$, and choose s to clear the stock. Young mean housing is at least $H/2$.

Allocation result. In the equilibrium on pages 3–4, $h_i^{y,E} = r$ and $H = r + m > 2r$. Consequently,

$$\boxed{\overline{h}^{y,F} \geq \frac{r+m}{2} > r = \overline{h}^{y,E}.}$$

The full planner allocates more housing to the young. With common fertility and slack caps, the increase for each young household is

$$h^{y,F} - h^{y,E} = \frac{m - r + \kappa n}{2} > 0.$$

Under heterogeneous fertility the aggregate claim still holds; an individual gain is checked from the displayed allocation, not inferred from its mean.

The comparison is financed by current transfers $q\Delta c_i + DP\Delta h_i$. They sum to zero. Changes in title holdings are offset by financial claims, including those of rental intermediaries, keeping young continuation wealth and old net estates unchanged. This is a dated utilitarian result, not a comparison that omits an initial old cohort.

6. Fertility: a condition without a restriction on βR. Now let the same planner choose young fertility, valuing current parents' utility. Children require goods and space in the young period only. Future parental utility therefore remains fixed under the same continuation-wealth settlement. No utility for unborn households is added.

Let n_0 be mean private fertility and $\overline{x}$ mean young adult consumption in an equilibrium where private fertility is optimal. Since $c_i^o = (\beta/q)x_i$, current goods per young-cohort unit are

$$\mathcal{C} = (1+\beta/q)\overline{x} + \chi n_0.$$

The joint planner chooses common fertility n^F, with adult consumption and young adult space

$$X(n) = \frac{\mathcal{C} - \chi n}{2}, \qquad S(n) = \min\left\{\frac{r+m-\kappa n}{2},\ H_O - \kappa n\right\}.$$

Its unique interior fertility choice satisfies

$$\boxed{\frac{\vartheta}{n^F} = \frac{\chi}{X(n^F)} + \frac{\alpha\kappa}{S(n^F)}.}$$

On either housing-cap regime this is a quadratic. Thus consumption and housing are reoptimized when the planner chooses fertility.

There is an exact test for a rise in mean fertility:

$$\boxed{n^F > n_0 \quad\Longleftrightarrow\quad \frac{\vartheta}{n_0} > \frac{2\chi}{(1+\beta/q)\overline{x}} + \frac{\alpha\kappa}{S(n_0)}.}$$

If $\beta \geq q$, the equilibrium's old consumption supplies a sufficient goods cushion and the inequality holds. When $\beta < q$, a simple sufficient condition, valid under income heterogeneity and binding planner caps, is

$$\boxed{\frac{\chi r}{\kappa\overline{x}}\frac{q-\beta}{q+\beta} < \alpha\frac{m-r}{m+r}.}$$

The left side measures the goods cost of supporting children when consumption is equalized across ages. The right side measures the available housing gain. Cheap child goods or a larger initial housing difference makes a fertility increase easier. This is not a condition on a credit multiplier.

A stronger condition that works for *every* $\beta > 0$ is

$$\frac{\chi r}{\kappa\overline{x}} < \alpha\frac{m-r}{m+r}.$$

In the explicit family, the right side is .2. Impose the primitive restrictions

$$g_i = \frac{w_i}{q} + Y_i, \qquad \left|\frac{g_i}{\bar{g}} - 1\right| \leq .01, \qquad \frac{\chi r}{\kappa} \leq .02\bar{g}, \qquad \bar{g} \geq 46\max_i w_i.$$

These primitive restrictions strengthen the family on page 3. For any specified $\nu > \kappa/r$, a unique scalar calibration of ϑ makes mean private fertility equal $1/\nu$ at the normalized cohort size. The price is then explicit and individual fertility is quadratic. The proof checks all fertility and tenure deviations. Throughout this family, $\chi r/(\kappa\bar{x}) < 1/12 < 1/5$, so the full joint planner raises mean fertility, including $\beta R < 1$. Calibrating ϑ to replacement is not a closed-form solution for ϑ or a claim of global equilibrium uniqueness.

7. The transition in the same model. Keep the entrant distribution of (w, Y) fixed and exogenous at every date; there is no endogenous transmission of types through fertility or estates. A permanent taste shock arrives when the stage opens. With adult household masses N_t^y and N_t^o, demographic accounting is

$$N_{t+1}^y = \nu \bar{n}_t N_t^y, \qquad N_{t+1}^o = N_t^y.$$

Both positive stationary endpoints have $\bar{n}^* = 1/\nu$, and population means refer to adult households, so a stationary total is $N^y + N^o = 2N$.

In this equilibrium, old households leave only housing in their estates. Write

$$a = \frac{\gamma + \omega_B}{1 + q\tau}, \qquad G = 1 + \beta K, \qquad C_0 = (1-q)\left(\frac{1}{q} + \frac{\tau}{2}\right), \qquad \widetilde{G} = G - \frac{(1+q)\tau a \beta}{2}.$$

At replacement fertility, the stationary endpoints satisfy

$$P^* = \frac{\bar{g} - \chi/\nu - \widetilde{G}\bar{x}^*}{C_0 r}, \qquad N^* = \frac{\bar{H}}{r + a\beta\bar{x}^*/P^*}.$$

Here $\bar{x}^*$ is the mean from aggregating individual quadratic fertility choices at replacement. Prices, rebates, and inherited saving are endogenous along the path.

The local theorem for the explicit endogenous family is as follows. A small permanent decline in ϑ lowers mean fertility on impact and lowers the endpoint N. At any finite date on that nearby baseline, a small unexpected permanent rebated-tax increase raises mean fertility on impact and raises the endpoint N. The path exists uniquely locally in the verified regime and converges to its nearby stationary endpoint. At the surprise, inherited saving is held fixed while the actual rebate changes: the current purchase price falls on impact, whereas the endpoint price rises. Young households continue to rent at $h = r$; the tax response works through goods, rents, saving, and rebates and does not implement the planner's full housing upgrade.

Mean inherited financial wealth is $\bar{a}_t$. Linear old demand in inherited saving yields an exact three-dimensional aggregate recursion in $(N_t^y, N_t^o, \bar{a}_t)$. The aggregate fertility elasticity, defined as the elasticity of mean fertility with respect to mean adult consumption, satisfies $\epsilon < .13$, and the root inequalities establish local stability in the verified regime. The full calculation supporting local convergence is in the supporting record. Fertility need not be higher at every subsequent date, and convergence need not be monotone.

8. What to keep, and what remains to do. The proposed cash-flow model gives the clearest combination obtained in this pass: two ages, solved household choices, a complete stationary example, the full dated housing result, and a fertility condition with an economic interpretation. Its substantive choices should be decided before changing the main deck:

1. Treat w as cash already received and Y as later working income. Consumption for both current ages and the old estate are delivered at period end. This is a change in financial timing, not a change of letters.

2. Let tenure change with age. The first illustration has no ownership taste; income and wealth still differ across households. The planner may reassign tenure and works with the original common divisible stock.

3. Use the same settlement ordering at both ages: pay current consumption and property tax from liquid receipts, then liquidate the house; young house-sale proceeds may repay the mortgage, and the next old-entry rebate arrives after settlement. Require pre-sale liquidity and post-settlement solvency. All young households rent in the proved family.

The older calculations remain useful alternatives:

Alternative	What it delivers	Main cost
Original timing, owner minimum, mobile tenure	Explicit fixed-fertility price and the intended age allocation	Its checked family still requires strong old-income and estate demand.
Three ages, with a mature earning period	An analytical equilibrium construction and a full planner result; quadratic or cubic household choices	An additional age and financial restriction; the useful family uses high mature income.
Original two-age log specification, weaker estate assumptions	An adult-space welfare result at conventional LTVs	In the checked moderate-taste family, old gross homes are much smaller than young homes.
Real moving costs	A reason for slow resizing after a fertility shock	The planner also pays those costs; retention alone does not prove misallocation.

More generally, concavity alone does not order housing by age. If $v(h,n)$ is housing utility, child-space increasing differences such as $v_h(h,n) > v_h(h,0)$ provide the relevant marginal preference ingredient; the log housing term satisfies it.

Remaining work. Global dynamics, larger shocks, endogenous type selection, and a policy implementing the housing upgrade remain unproved.

Relation to Coven et al. Their August 2026 lifecycle model includes a minimum owner size and separate rental and owner stocks. Their two-period illustration studies housing as an asset without housing utility. It therefore does not supply the dated housing-utility theorem sought here. The proposed common-stock model is a separate illustrative exercise.[1]

[1] Joshua Coven, Sebastian Golder, Arpit Gupta and Abdoulaye Ndiaye, *Property Taxes and Housing Allocation Under Financial Constraints*, August 1, 2026, pp. 15, 20–23. `https://abdouecon.github.io/research/papers/Property_Tax.pdf`